

ABDUR RAFAY

AI Engineer | Voice Agents | Multi-Agent Systems | DevOps
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PROFESSIONAL SUMMARY

Member of Technical Staff at Lythe Labs (SF), building and shipping production voice agents, multi-agent systems, and the infrastructure that runs them. Own CI/CD pipelines, AWS production stack, and MCP server development end-to-end. Comfortable going from system design to deployment solo across distributed remote teams. Currently completing a BS in Artificial Intelligence at Bahria University.

PROFESSIONAL EXPERIENCE

Member of Technical Staff | Lythe Labs | San Francisco (Remote)

Mar 2026 – Present

- Designed and shipped a production-ready Voice Agent Starter Kit enabling natural phone-call-style LLM interactions, cutting voice agent integration time for new projects.
- Built voice pipelines and tool-calling agent infrastructure, working across the full stack from LLM integration to telephony-side tooling.
- Engineered a long-running autonomous agent supporting weeks of continuous runtime using LLM planning, sandboxed code execution, persistent memory, and resumable sessions.
- Built Agent Testing Suite dashboard tracking latency, API cost, and tool call frequency with interactive flow visualization for multi-agent debugging.
- Integrated Jira and Linear for issue tracking and defect triage across voice pipeline and agent system development cycles.
- Designed and maintained test plans for voice pipelines and agent systems; triaged failures across telephony-side tooling and LLM integrations.
- Owned CI/CD infrastructure with automated test integration and communicated design decisions across a distributed remote team.

AI Intern | School of AI | Georgia, USA (Remote)

Jan 2026 – Mar 2026

- Developed 25+ AI agent systems independently across healthcare, finance, cybersecurity, and education; delivered each project end-to-end with minimal supervision.
- Built full-stack multi-agent platforms featuring persistent memory, voice control, workflow automation, and document analysis, reducing manual document processing time by an estimated 70% in pilot tests.
- Implemented agent-to-agent communication logic, context-aware task coordination, and AI-driven structured outputs using LangChain and Python.
- Key systems shipped: medical note structurer (structured 100% of unformatted clinical notes into standard templates), earnings call analyzer, incident response AI assistant, AI tutoring platform, and collaborative document analyzer.
- Communicated system designs and progress weekly to cross-functional team leads across 3 time zones.

AI Intern | AABS | Lahore, Pakistan

Nov 2025 – Jan 2026

- Built and optimized production-ready RAG pipelines using Groq API and ChromaDB, achieving low-latency LLM responses at scale with accurate document retrieval.
- Developed PDF chatbot systems and memory-augmented conversational agents; improved retrieval precision by 40% through embedding pipeline tuning and relevance optimization.
- Integrated SQL databases (PostgreSQL) for session and metadata storage in multi-agent MLOps workflows.
- Applied user-intent classification logic and multi-agent orchestration to route queries accurately, reducing irrelevant LLM responses.

Machine Learning Intern | Elevvo Pathways | Cairo, Egypt (Remote)

Sep 2025

- Delivered end-to-end ML solutions independently: Linear Regression, Logistic Regression, K-Means clustering, CNNs, XGBoost, and collaborative filtering recommendation system.
- Executed the full project lifecycle (data cleaning, feature engineering, model training, evaluation, and visualization) across 6 distinct ML projects in one month.

Data Analyst Intern | Excelerate | Dubai, UAE (Remote)

Aug 2025 – Sep 2025

- Led a team of 4 interns analyzing social media marketing data; identified 3 underperforming ad campaigns and presented cost-reduction recommendations adopted by the finance team.
- Built interactive dashboards in Google Looker Studio using Pandas and NumPy; wrote SQL queries to extract and aggregate campaign performance data across multiple sources.

Data Science Intern | Securely Innovations (Pvt.) Ltd. | Islamabad, Pakistan

Sep 2024 – Nov 2024

- Preprocessed and analyzed large structured and unstructured datasets using Pandas, NumPy, and SQL databases; automated reporting workflows that cut manual processing time by 30%.
- Applied statistical modeling and data mining techniques to surface actionable patterns for the security analytics team.
- Conducted QA testing across the company's main web platform including functional, regression, and mobile responsiveness testing; tracked and triaged defects using Jira.

EDUCATION

Bachelor of Science in Artificial Intelligence

2024 – Present

Bahria University | Islamabad, Pakistan

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, C++

Databases: PostgreSQL, MySQL, ChromaDB (vector)

ML / Deep Learning: TensorFlow, PyTorch, Scikit-learn, XGBoost, Hugging Face, Sentence Transformers

LLM & AI: LangChain, LLMs (OpenAI, Anthropic Claude, Groq, Google Gemini), RAG, multi-agent systems

Backend & DevOps: FastAPI, Flask, Streamlit, Docker, Git, CI/CD

QA & Testing: pytest, API testing, regression testing, CI/CD test integration, defect triage, Jira, Linear, test case design

Data & Analytics: Pandas, NumPy, Matplotlib, Google Looker Studio

Soft Skills: Independent work, cross-functional communication, remote team collaboration, technical documentation

PROJECTS

Relay - Arclat | github.com/arclat-ai/relay-mcp

MCP server that audits Python voice agent codebases for async latency bottlenecks. Detects blocking calls, unparallelized awaits, and inefficient tool chains at the AST level. Published to PyPI as relay-arclat.

Crucible | github.com/Abdur-Rafay-AR/crucible

Multi-agent framework with hybrid BM25 + embedding SQLite memory and local Ollama support. Zero core dependencies. Built for reproducible, self-hosted agent workflows.

Agent Testing Suite - Lythe Labs | <https://github.com/lythelab/agent-testing-suite/>

Evaluation dashboard tracking latency, API cost, and tool call frequency across agent runs. Interactive flow visualization for debugging multi-agent pipelines.

ai-cache | <https://github.com/Abdur-Rafay-AR/ai-cache>

Open-source Python library for automatic LLM API response caching across OpenAI, Anthropic, and Gemini. Reduces costs and latency for repeated NLP queries.

CERTIFICATIONS

- [Generative AI with Large Language Models](#) - Amazon Web Services / DeepLearning.AI
- [AI Infrastructure and Operations Fundamentals](#) - NVIDIA
- [Supervised Machine Learning: Regression and Classification](#) - Stanford University / DeepLearning.AI
- [AI Developer Professional Certificate](#) - IBM